IoT-Based Air Quality Monitoring System

A internship report submitted in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted By

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Under the esteemed guidance of

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(Deemed to be University)
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2018 – 2022

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Place: Visakhapatnam

Abstract

Air pollution has become one of the major environmental concerns that affect human health and quality of life. Continuous monitoring of air quality is essential to identify harmful pollutants and create awareness about environmental conditions. The objective of this project is to develop an IoT-Based Air Quality Monitoring System capable of monitoring air quality in real time and providing alerts when pollution levels exceed safe limits.

The system utilizes an MQ gas sensor interfaced with an Arduino Uno microcontroller to detect the concentration of harmful gases present in the atmosphere. The sensor readings are continuously monitored and analyzed by the microcontroller. Based on the measured air quality values, the system classifies the environment into three categories: Fresh Air, Poor Air, and Very Poor Air. The status is displayed on a 16×2 LCD module for easy user interpretation.

To enhance user awareness and safety, visual and audible alerts are incorporated into the system. A green LED indicates fresh air conditions, while a red LED indicates deteriorating air quality. When the air quality reaches a critical level, a buzzer is activated and a warning message is displayed on the LCD, advising users to avoid going outdoors unless necessary.

The proposed system is simple, cost-effective, and suitable for deployment in homes, educational institutions, offices, and industrial environments. It demonstrates the practical application of embedded systems and Internet of Things (IoT) concepts in environmental monitoring and public health protection.

Tanuja Dasari

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Chapter 1

Introduction

Air pollution is one of the most significant environmental challenges faced by modern society. Rapid industrialization, urbanization, increasing vehicular emissions, and the extensive use of fossil fuels have contributed to the deterioration of air quality worldwide. Poor air quality not only affects the environment but also poses serious health risks to human beings, leading to respiratory diseases, cardiovascular disorders, and other health complications.

Continuous monitoring of air quality is essential for assessing pollution levels and taking timely preventive measures. Traditional air quality monitoring systems are often expensive and require sophisticated infrastructure, making them less accessible for widespread deployment. Recent advancements in embedded systems and Internet of Things (IoT) technologies have enabled the development of cost-effective and portable environmental monitoring solutions.

The IoT-Based Air Quality Monitoring System is designed to monitor air quality in real time using gas sensors and a microcontroller-based platform. The system detects the concentration of harmful gases present in the surrounding atmosphere and categorizes the air quality into different levels such as Fresh Air, Poor Air, and Very Poor Air. The monitored data is displayed on an LCD screen, providing users with immediate information about environmental conditions.

To enhance public awareness and safety, the system incorporates visual and audible alert mechanisms. A green LED indicates healthy air conditions, while a red LED warns users about deteriorating air quality. When pollution levels exceed a predefined threshold, a buzzer is activated and a warning message is displayed, advising users to avoid outdoor activities unless necessary.

The proposed system demonstrates the practical application of embedded systems and IoT concepts in environmental monitoring. It offers a simple, affordable, and efficient solution that can be deployed in homes, offices, educational institutions, and industrial environments to help individuals make informed decisions regarding their exposure to air pollution.

This project highlights the importance of technology-driven environmental monitoring systems and contributes towards creating healthier and safer living conditions through real-time air quality assessment.

Chapter 2

Project Work

This chapter presents the complete development process of the IoT-Based Air Quality Monitoring System. It discusses the inspiration behind the project, objectives, design methodology, literature review, hardware and software requirements, system architecture, circuit implementation, and operational workflow. The project aims to provide a cost-effective solution for monitoring air quality and alerting users when pollution levels become hazardous.

The system is developed using an Arduino Uno microcontroller, an MQ gas sensor, a 16×2 LCD display, LEDs, and a buzzer. The gas sensor continuously monitors the concentration of harmful gases in the environment. Based on the sensor readings, the system classifies air quality into Fresh Air, Poor Air, and Very Poor Air categories. Appropriate visual and audible indications are provided to inform users about the current environmental conditions.

The following sections describe the various stages involved in the design and implementation of the proposed system.

2.1 Inspiration

Air pollution has emerged as one of the most critical environmental and public health challenges in recent years. Rapid urbanization, industrial growth, and increasing vehicular emissions have significantly contributed to the degradation of air quality. Exposure to polluted air can lead to various health problems, including respiratory diseases, allergies, asthma, and cardiovascular disorders.

In many cities, people are often unaware of the quality of air they breathe daily. While government agencies operate large-scale air quality monitoring stations, these systems are expensive and limited in number. As a result, real-time air quality information is not always readily available at every location. This creates a need for affordable, portable, and user-friendly air quality monitoring solutions.

The inspiration for this project came from the growing concern about environmental pollution and the need to create awareness regarding air quality. By leveraging the capabilities of embedded systems and IoT technologies, it is possible to develop a low-cost system capable of continuously monitoring air quality and alerting users whenever pollution levels become unsafe.

The proposed IoT-Based Air Quality Monitoring System aims to provide individuals with real-time information about their surrounding environment. The system not only monitors harmful gases but also classifies the air quality into different categories and provides visual and audible alerts. Such a solution can help people take necessary precautions and contribute towards promoting a healthier and safer environment.

2.2 Project Objectives

The primary objective of the IoT-Based Air Quality Monitoring System is to monitor environmental air quality in real time and alert users whenever pollution levels exceed safe limits.

The specific objectives of the project are:

- To design and develop a low-cost air quality monitoring system using Arduino Uno.
- To measure the concentration of harmful gases using an MQ gas sensor.
- To classify air quality into Fresh Air, Poor Air, and Very Poor Air categories.
- To display air quality status and sensor readings on a 16×2 LCD display.
- To provide visual indications using LEDs for different air quality levels.
- To generate an audible alert through a buzzer when air quality becomes hazardous.
- To create an easy-to-use and portable solution suitable for homes, offices, and educational institutions.
- To demonstrate the practical application of IoT and embedded systems in environmental monitoring.

2.3 Design Idea

The proposed IoT-Based Air Quality Monitoring System is designed to continuously monitor the concentration of harmful gases present in the atmosphere and provide immediate feedback to users regarding air quality conditions. The system primarily consists of an Arduino Uno microcontroller, an MQ gas sensor, a 16×2 LCD display, LEDs, and a buzzer.

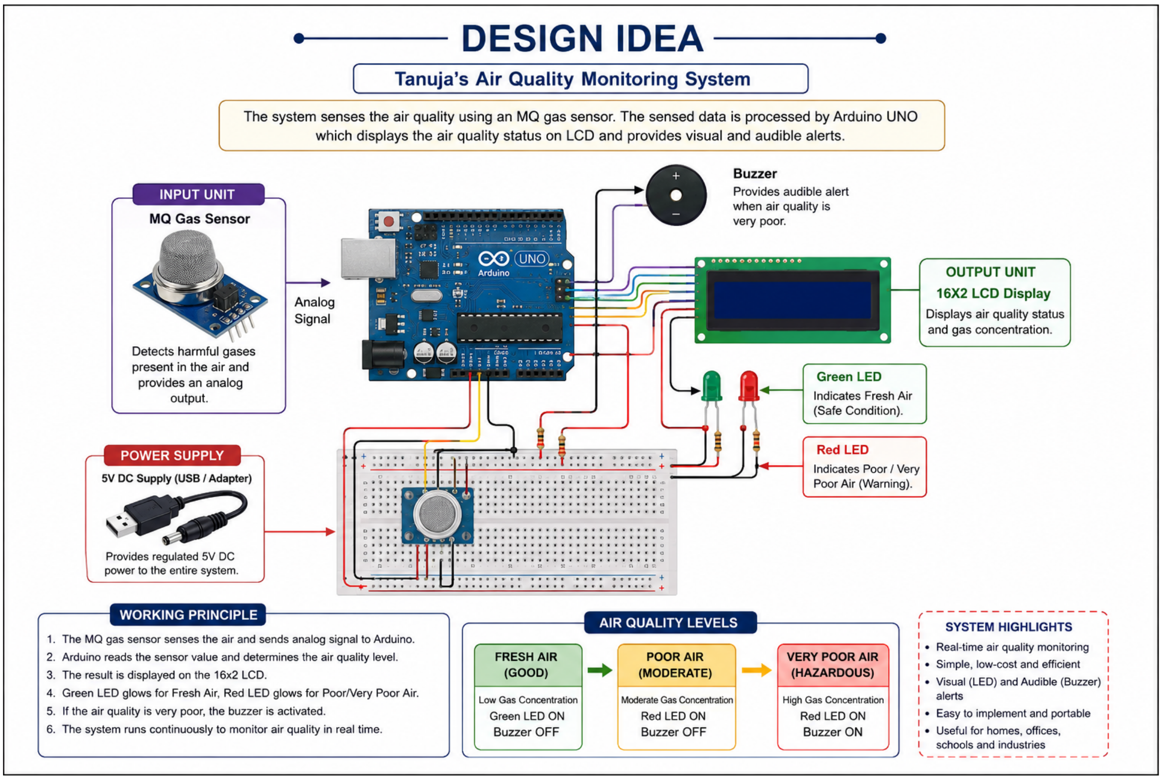
The MQ gas sensor continuously senses the surrounding air and generates an analog signal proportional to the concentration of gases present. The Arduino Uno reads these sensor values through its analog input pin and processes the data according to predefined threshold levels.

Based on the sensor readings, the air quality is classified into three categories:

- Fresh Air
- Poor Air
- Very Poor Air

For Fresh Air conditions, the green LED is turned ON, indicating a safe environment. When the air quality deteriorates to the Poor Air category, the red LED is activated to alert users. If the pollution level exceeds the critical threshold and falls into the Very Poor Air category, the buzzer generates an alarm and a warning message is displayed on the LCD advising users to avoid going outdoors unless necessary.

The LCD module provides real-time information about the current air quality status and sensor readings. The system is designed to be simple, cost-effective, and easy to deploy in residential, educational, and workplace environments for continuous environmental monitoring.



2.4 Literature Review

Air quality monitoring has become an important area of research due to the increasing impact of environmental pollution on human health and ecological balance. Several researchers have developed monitoring systems using sensor networks, wireless communication technologies, and Internet of Things (IoT) platforms to collect and analyze environmental data in real time.

Traditional air quality monitoring stations provide highly accurate measurements; however, they are expensive to install and maintain. Their limited deployment often restricts coverage to specific urban locations. To overcome these limitations, researchers have proposed low-cost monitoring systems using microcontrollers and gas sensors that can provide localized air quality information.

Recent studies have demonstrated the effectiveness of IoT-based air quality monitoring systems in collecting environmental data and transmitting it to cloud platforms for analysis. These systems typically utilize gas sensors such as MQ-series sensors, temperature sensors, humidity sensors, and wireless communication modules. The collected data can be displayed locally or transmitted remotely for further processing and visualization.

Arduino-based monitoring systems have gained popularity due to their affordability, ease of implementation, and compatibility with a wide range of sensors. Such systems enable continuous monitoring of pollutants and provide timely alerts whenever pollution levels exceed safe limits. The integration of LCD displays, LEDs, and alarm mechanisms further improves user awareness and safety.

The proposed IoT-Based Air Quality Monitoring System builds upon these concepts by providing a simple and economical solution for real-time air quality assessment. The system focuses on monitoring harmful gases, classifying air quality levels, and generating appropriate alerts to help users make informed decisions regarding environmental exposure.

The literature survey indicates that low-cost embedded systems can effectively contribute to environmental monitoring and public health protection while remaining accessible for educational and practical applications.

2.5 System Design Requirements

The proposed IoT-Based Air Quality Monitoring System consists of various hardware components that work together to monitor environmental air quality and provide real-time alerts. Each component plays a specific role in sensing, processing, displaying, and alerting users about air quality conditions.

2.5.1 MQ Gas Sensor

The MQ Gas Sensor is the primary sensing component of the system. It is used to detect the concentration of harmful gases and pollutants present in the surrounding atmosphere. The sensor produces an analog voltage proportional to the gas concentration, which is read by the Arduino Uno through its analog input pin.

The MQ sensor is widely used because of its low cost, high sensitivity, and ease of interfacing with microcontrollers. In this project, the sensor continuously monitors air quality and provides data for classification into Fresh Air, Poor Air, and Very Poor Air categories.

Specifications:

- Operating Voltage: 5V
- Output Type: Analog
- Gas Detection: Smoke and Harmful Gases
- Response Time: Fast
- Interface: Analog Output Pin

MQ Gas Sensor

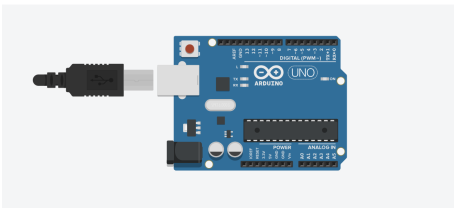
2.5.2 Arduino UNO

Arduino Uno serves as the central processing unit of the system. It receives sensor readings from the MQ Gas Sensor, processes the data, and controls the LCD display, LEDs, and buzzer based on predefined threshold values.

Arduino Uno is based on the ATmega328P microcontroller and is widely used in embedded system applications due to its simplicity, flexibility, and large community support.

Specifications:

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Digital I/O Pins: 14
- Analog Input Pins: 6
- Clock Speed: 16 MHz



Arduino UNO

2.5.3 16×2 LCD Display

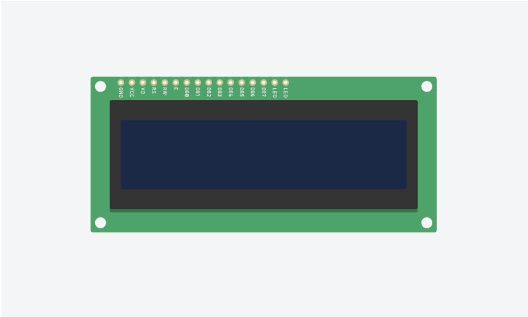
The 16×2 LCD Display is used to present air quality information to the user. It displays the current air quality status along with the sensor reading in real time.

The LCD operates in 4-bit mode to reduce the number of Arduino pins required for interfacing.

Specifications:

- Display Type: 16×2 Character LCD
- Operating Voltage: 5V
- Display Capacity: 32 Characters

- Interface Mode: 4-bit



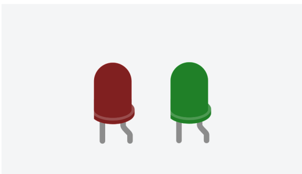
16×2 LCD Display

2.5.4 LEDs

Two LEDs are used as visual indicators for air quality status. The Green LED indicates Fresh Air conditions, while the Red LED indicates Poor Air or Very Poor Air conditions. These indicators provide a quick visual representation of environmental conditions.

Specifications:

- Operating Voltage: 2V–3V
- Current Consumption: 20mA
- Colors Used: Green and Red



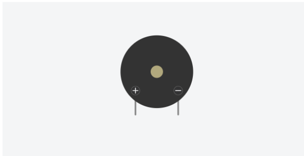
Green and Red LEDs

2.5.5 Buzzer

The buzzer provides an audible warning whenever the air quality reaches the Very Poor Air category. This helps users become aware of hazardous conditions even when they are not looking at the display.

Specifications:

- Operating Voltage: 5V
- Sound Output: Audible Tone
- Control Type: Digital



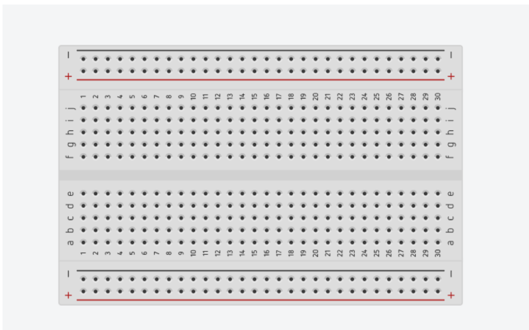
Buzzer

2.5.6 Breadboard and Jumper Wires

The breadboard and jumper wires are used to establish temporary electrical connections among all hardware components without soldering. They simplify circuit assembly and testing during prototype development.

Specifications:

- Solderless Prototyping
- Reusable Design Platform
- Easy Circuit Modification



Breadboard and Jumper Wires

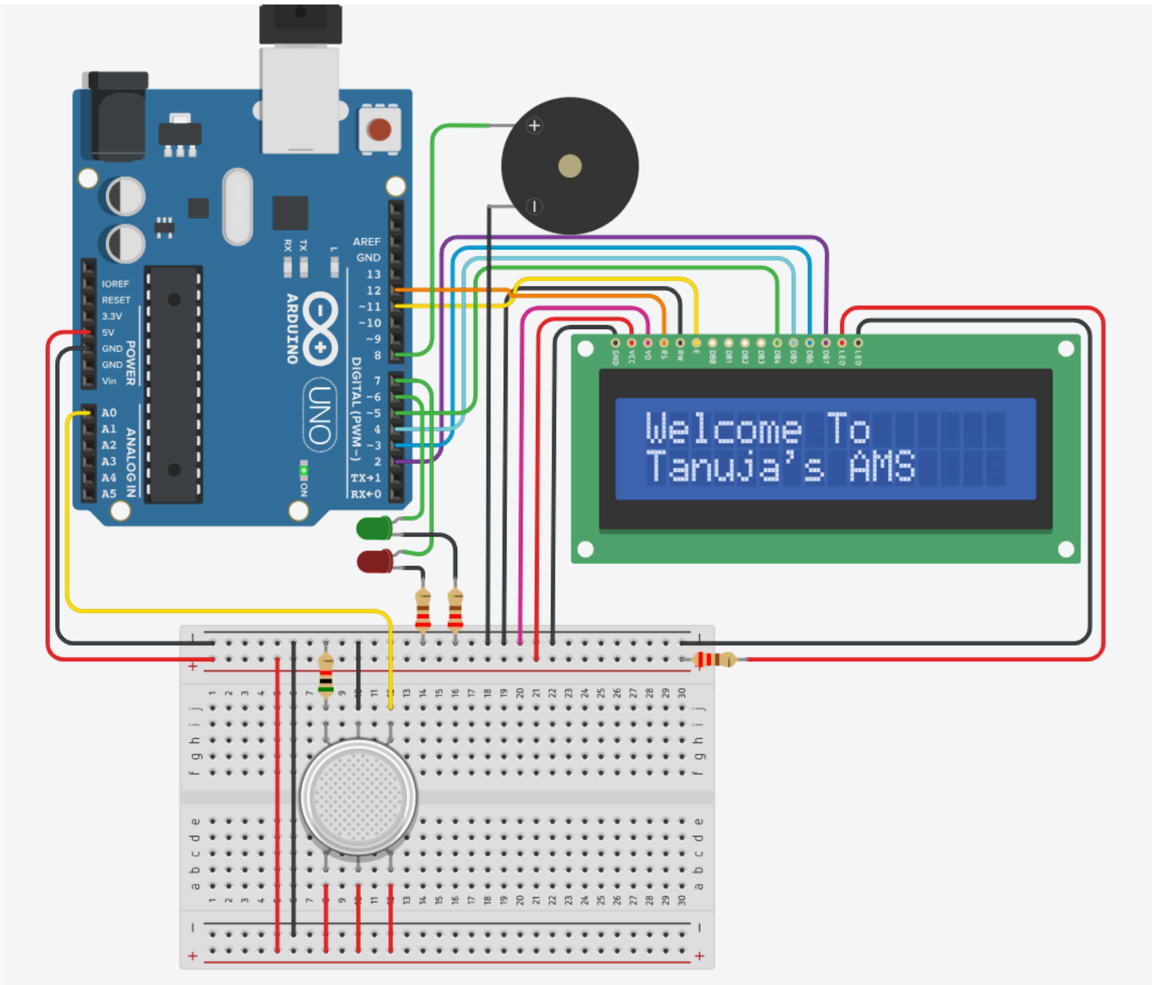
2.6 Block Diagram

The block diagram illustrates the overall architecture and working flow of the proposed IoT-Based Air Quality Monitoring System. The MQ Gas Sensor continuously monitors the concentration of harmful gases present in the surrounding environment and sends the sensed data to the Arduino Uno in the form of analog signals.

The Arduino Uno acts as the central processing unit of the system. It reads and analyzes the sensor values and classifies the air quality into three categories namely Fresh Air, Poor Air, and Very Poor Air based on predefined threshold values.

The processed information is displayed on the 16×2 LCD Display to provide real-time air quality status to the user. Visual indications are provided using Green and Red LEDs. The Green LED glows when the air quality is good, whereas the Red LED glows when the air quality deteriorates.

Whenever the air quality reaches the Very Poor Air category, the Arduino activates the buzzer to alert users about hazardous environmental conditions. Thus, the system provides both visual and audible notifications for effective air quality monitoring.



Block Diagram of IoT-Based Air Quality Monitoring System

2.7 Working

The IoT-Based Air Quality Monitoring System continuously monitors the surrounding air for the presence of harmful gases using an MQ Gas Sensor. The sensor produces an analog output voltage that varies according to the concentration of gases present in the environment. This analog signal is sent to the Arduino Uno through analog pin A0.

Initially, when the system is powered ON, a welcome message 'Welcome To Tanuja's AMS" is displayed on the 16×2 LCD screen, followed by the project title 'Air Monitoring System." This indicates that the system has started successfully and is ready to monitor the air quality.

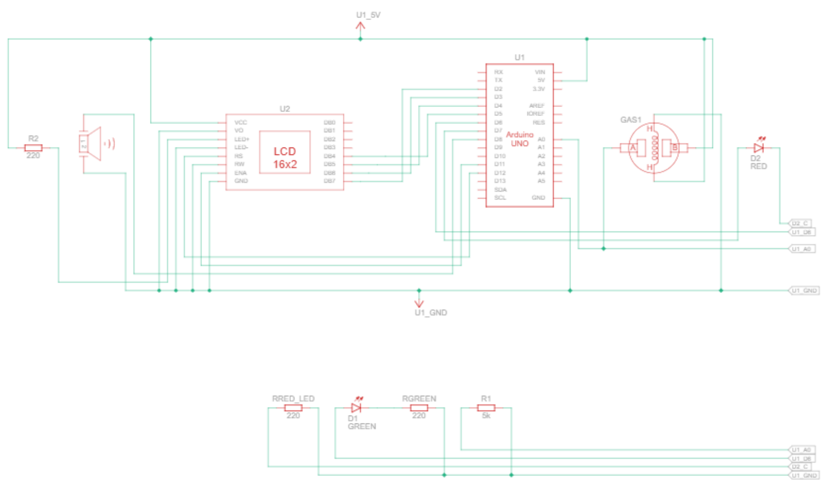
The Arduino Uno continuously reads the analog values generated by the MQ Gas Sensor using the `analogRead()` function. These sensor values are processed and compared with predefined threshold values to determine the current air quality.

Based on the sensor readings, the air quality is classified into three categories:

- **Fresh Air:** When the sensor value is less than 350, the air is considered clean. The Green LED is turned ON, the Red LED remains OFF, and the buzzer remains silent. The LCD displays “Fresh Air” along with the current air quality value.
- **Poor Air:** When the sensor value is between 350 and 699, the air quality is considered poor. The Green LED is turned OFF, while the Red LED is turned ON to indicate caution. The buzzer remains OFF, and the LCD displays “Poor Air” together with the sensor reading.
- **Very Poor Air:** When the sensor value is 700 or above, the air quality becomes hazardous. The Red LED remains ON and the buzzer generates an audible alarm to warn the user. The LCD displays the warning message “Avoid Going Out” along with the current air quality value, advising users to avoid outdoor activities unless absolutely necessary.

The system continuously repeats this process by monitoring the surrounding air in real time. Any change in the gas concentration is immediately reflected on the LCD display, LEDs, and buzzer. This enables users to receive instant notifications regarding environmental conditions and take appropriate preventive measures.

The overall working of the proposed system is illustrated in the flowchart shown below.



Working Flow of the IoT-Based Air Quality Monitoring System

Chapter 3

Source Code

This chapter presents the software implementation of the IoT-Based Air Quality Monitoring System. The program was developed using the Arduino Integrated Development Environment (IDE) in Embedded C/C++. The Arduino Uno continuously reads data from the MQ Gas Sensor, processes the sensor values, classifies the air quality, and controls the LCD display, LEDs, and buzzer accordingly.

The source code consists of initialization routines, sensor data acquisition, decision-making logic, and output control. The complete Arduino program used for implementing the proposed system is presented below.

3.1 Software Development

The software for the proposed system was developed using the Arduino IDE. The Arduino programming language, which is based on C/C++, was used to implement the logic for monitoring air quality.

The program initializes the LCD display, LEDs, buzzer, and gas sensor during system startup. It continuously acquires analog data from the MQ Gas Sensor through the Arduino's analog input pin. Based on predefined threshold values, the program determines the current air quality category and activates the corresponding output devices.

The software follows a simple and efficient algorithm that enables real-time monitoring while ensuring low computational complexity and quick response to changes in air quality.

3.2 Arduino Program

The complete Arduino source code developed for the IoT-Based Air Quality Monitoring System is shown below.

Arduino Program for Air Quality Monitoring System

```

#include <LiquidCrystal.h>

// LCD: RS, EN, D4, D5, D6, D7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

```

```
int sensor = A0;
int airValue = 0;

// LEDs
int greenLED = 6;
int redLED   = 7;

// Buzzer
int buzzer = 8;

void setup()
{
  Serial.begin(9600);

  pinMode(greenLED, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(buzzer, OUTPUT);

  lcd.begin(16, 2);

  // Welcome Screen
  lcd.setCursor(0, 0);
  lcd.print("Welcome To");

  lcd.setCursor(0, 1);
  lcd.print("Tanuja's AMS");

  delay(3000);

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Air Monitoring");

  lcd.setCursor(0, 1);
  lcd.print("System");

  delay(2000);

  lcd.clear();
}

void loop()
{
  airValue = analogRead(sensor);

  Serial.print("Air Quality Value: ");
  Serial.println(airValue);

  // FRESH AIR
  if (airValue < 350)
```

```
  {
    digitalWrite(greenLED, HIGH);
    digitalWrite(redLED, LOW);
    noTone(buzzer);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Fresh Air");

    lcd.setCursor(0, 1);
    lcd.print("AQ:");
    lcd.print(airValue);
  }

  // POOR AIR
  else if (airValue < 700)
  {
    digitalWrite(greenLED, LOW);
    digitalWrite(redLED, HIGH);
    noTone(buzzer);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Poor Air");

    lcd.setCursor(0, 1);
    lcd.print("AQ:");
    lcd.print(airValue);
  }

  // VERY POOR AIR
  else
  {
    digitalWrite(greenLED, LOW);
    digitalWrite(redLED, HIGH);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Avoid Going Out");

    lcd.setCursor(0, 1);
    lcd.print("AQ:");
    lcd.print(airValue);

    tone(buzzer, 1000);
    delay(300);
    noTone(buzzer);
  }

  delay(1000);
}
```

3.3 Code Explanation

The Arduino program consists of two major functions namely `setup()` and `loop()`.

The `setup()` function executes only once after the Arduino is powered on. It initializes the serial communication, configures the LEDs and buzzer as output devices, initializes the 16×2 LCD display, and displays a welcome message indicating that the Air Monitoring System has started successfully.

The `loop()` function executes continuously throughout the operation of the system. It reads the analog value generated by the MQ Gas Sensor using the `analogRead()` function. The obtained sensor value is compared against predefined threshold values to classify the surrounding air quality.

When the sensor value is less than 350, the air is classified as Fresh Air. The Green LED glows, the Red LED remains OFF, and the buzzer is disabled. The LCD displays the message ”Fresh Air” along with the current air quality value.

When the sensor value lies between 350 and 699, the air quality is classified as Poor Air. The Red LED glows while the Green LED remains OFF. The LCD displays ”Poor Air” together with the sensor reading, and the buzzer remains inactive.

If the sensor value reaches or exceeds 700, the air quality is classified as Very Poor Air. The Red LED remains ON, the buzzer generates an audible warning, and the LCD displays the message ”Avoid Going Out” along with the measured air quality value. The alarm continues to alert users whenever hazardous air conditions are detected.

The entire monitoring process repeats every second, thereby providing continuous real-time air quality monitoring.

Chapter 4

Results and Discussion

This chapter presents the simulation results obtained from the implementation of the IoT-Based Air Quality Monitoring System using Tinkercad. The system was tested under different air quality conditions by varying the concentration of gases detected by the MQ Gas Sensor. Based on the sensor readings, the Arduino Uno successfully classified the air quality into three categories: Fresh Air, Poor Air, and Very Poor Air.

The corresponding outputs were displayed on the LCD screen, while the LEDs and buzzer responded according to the predefined threshold values. The experimental results confirm that the proposed system accurately monitors air quality and provides timely visual and audible alerts whenever hazardous conditions are detected.

4.1 Simulation Results

The proposed system was successfully simulated using the Tinkercad simulation platform. Various air quality conditions were created by changing the gas concentration around the MQ Gas Sensor. The Arduino continuously processed the sensor readings and generated appropriate outputs.

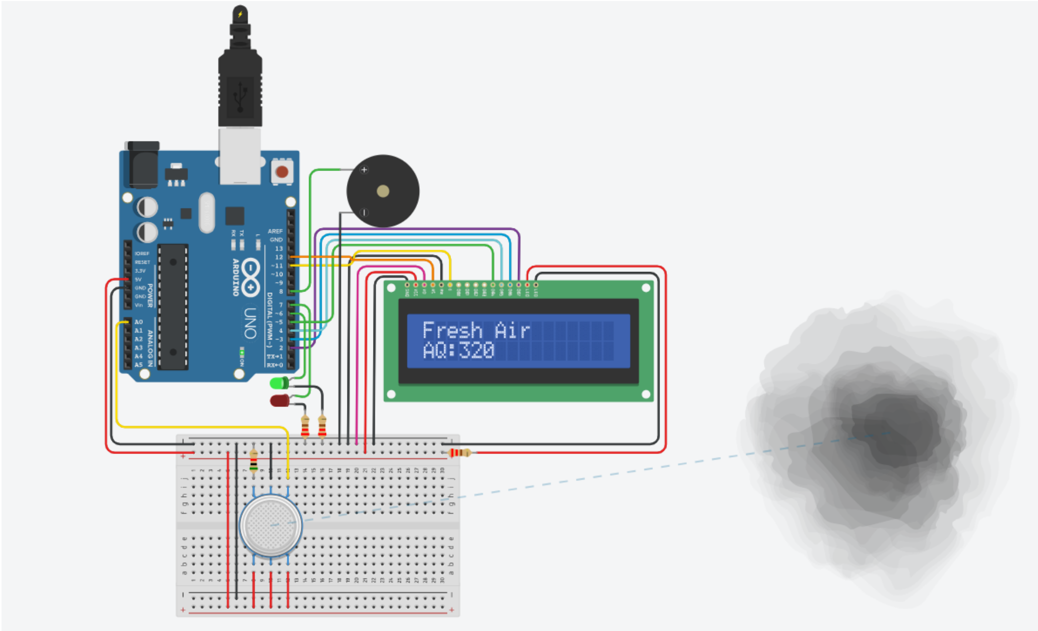
The LCD displayed the air quality status in real time, while the LEDs and buzzer responded according to the programmed logic. The obtained results verified the correct functioning of the hardware connections as well as the Arduino program.

4.2 Fresh Air Output

When the MQ Gas Sensor detects a low concentration of harmful gases (sensor value less than 350), the system classifies the surrounding environment as Fresh Air.

Under this condition:

- Green LED turns ON.
- Red LED remains OFF.
- Buzzer remains OFF.
- LCD displays ”Fresh Air” along with the air quality value.



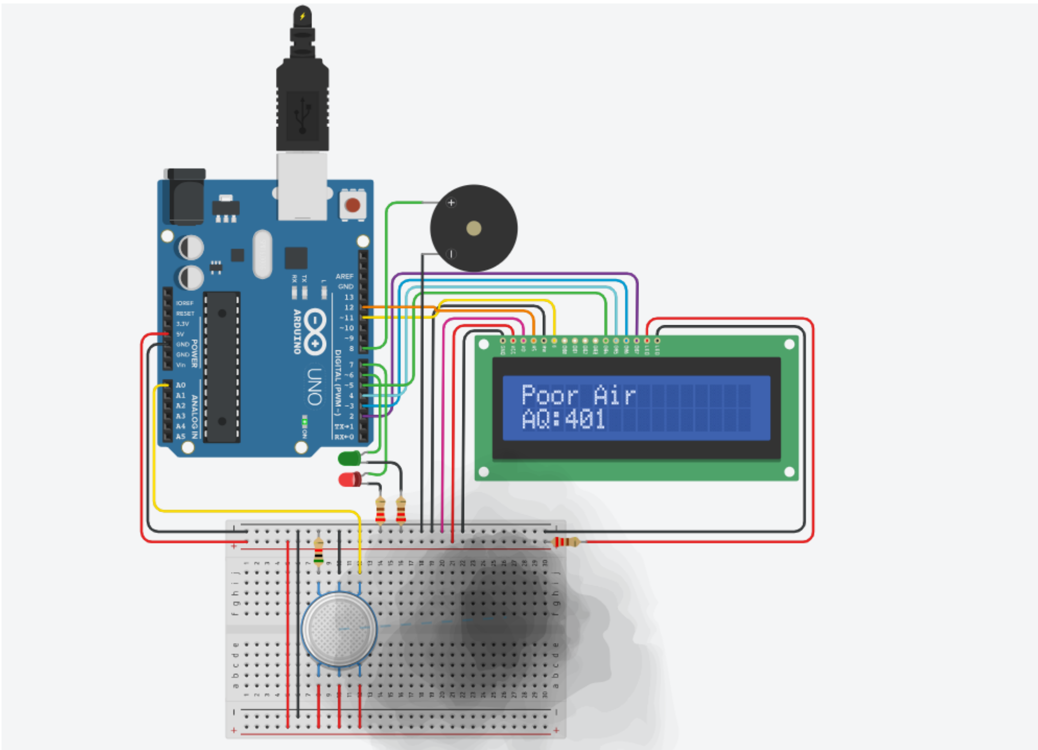
Fresh Air Output

4.3 Poor Air Output

When the MQ Gas Sensor detects a moderate concentration of harmful gases (sensor value between 350 and 699), the system categorizes the air as Poor Air.

During this condition:

- Green LED turns OFF.
- Red LED turns ON.
- Buzzer remains OFF.
- LCD displays ”Poor Air” along with the air quality value.



Poor Air Output

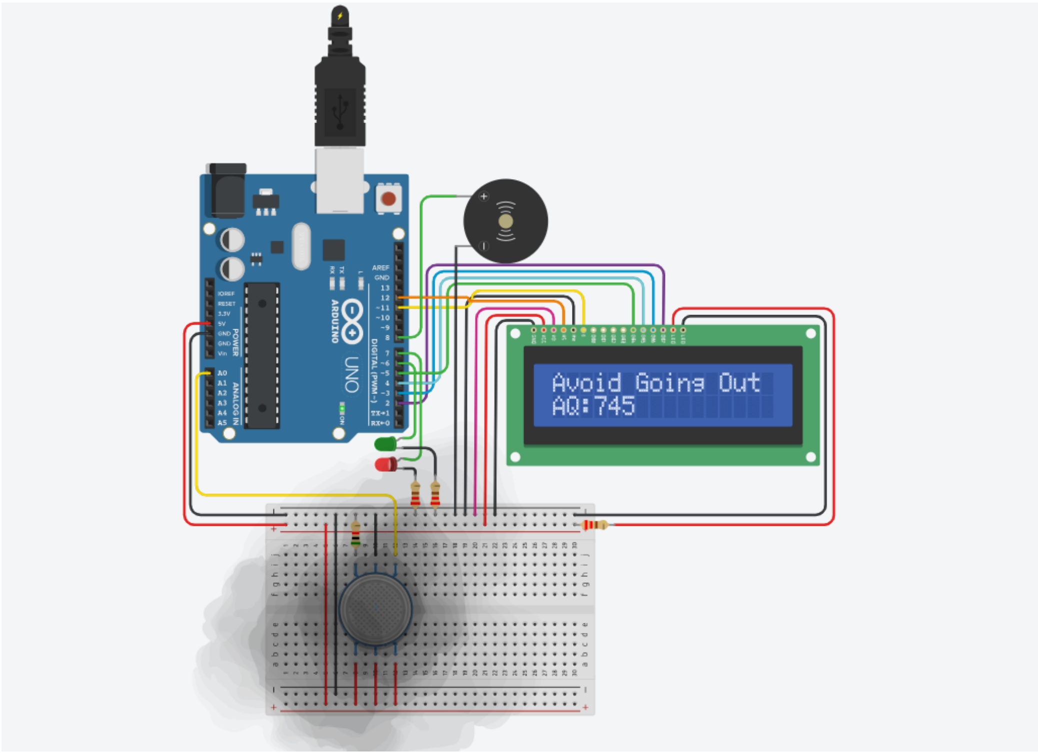
4.4 Very Poor Air Output

When the MQ Gas Sensor detects a high concentration of harmful gases (sensor value greater than or equal to 700), the system classifies the air as Very Poor Air.

Under hazardous conditions:

- Green LED remains OFF.
- Red LED turns ON.
- Buzzer generates an audible warning.
- LCD displays the warning message ”Avoid Going Out” together with the air quality value.

This alert mechanism helps users become aware of unsafe environmental conditions and encourages them to avoid unnecessary outdoor activities.



Very Poor Air Output

4.5 Discussion

The simulation results demonstrate that the proposed IoT-Based Air Quality Monitoring System performs efficiently under different air quality conditions. The MQ Gas Sensor accurately detects changes in gas concentration, and the Arduino Uno successfully processes the sensor data to classify the air quality.

The LCD provides real-time information regarding the current air quality, while the LEDs offer clear visual indications of environmental conditions. The buzzer effectively alerts users whenever the detected air quality reaches hazardous levels.

The system is simple, economical, and easy to implement, making it suitable for educational purposes as well as small-scale environmental monitoring applications. Although the project was simulated using Tinkercad, the same hardware configuration can be implemented using physical components with minimal modifications.

Chapter 5

Applications and Advantages

5.1 Applications

The IoT-Based Air Quality Monitoring System can be utilized in various environments where continuous monitoring of air quality is essential. The system provides real-time information about harmful gas concentration and helps users take preventive measures against air pollution.

Some important applications of the proposed system are:

- Indoor air quality monitoring in homes and apartments.
- Environmental monitoring in schools, colleges, and universities.
- Air quality monitoring in offices and commercial buildings.
- Pollution monitoring in industrial areas and factories.
- Monitoring of laboratories where harmful gases are present.
- Smart city environmental monitoring systems.
- Public health awareness regarding air pollution.
- Research and educational projects related to IoT and Embedded Systems.

5.2 Advantages

The proposed Air Quality Monitoring System offers several advantages over conventional monitoring methods.

- Low-cost implementation using readily available components.
- Simple circuit design and easy installation.
- Real-time monitoring of surrounding air quality.
- Instant visual indication through LEDs.
- Audible warning using a buzzer during hazardous conditions.

- User-friendly LCD display for easy interpretation of air quality.
- Low power consumption.
- Easily expandable by integrating additional environmental sensors.
- Suitable for educational, domestic, and industrial applications.

5.3 Limitations

Although the proposed system performs efficiently for basic air quality monitoring, it has certain limitations.

- The MQ Gas Sensor provides relative gas concentration values rather than precise pollutant measurements in parts per million (PPM).
- The system detects the presence of harmful gases but does not distinguish between different gas types.
- Sensor readings may vary depending on temperature, humidity, and calibration conditions.
- The project was validated through Tinkercad simulation; practical implementation may require sensor calibration for improved accuracy.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The IoT-Based Air Quality Monitoring System was successfully designed and simulated using Arduino Uno, an MQ Gas Sensor, a 16×2 LCD Display, LEDs, and a buzzer. The proposed system continuously monitors air quality and classifies it into three categories: Fresh Air, Poor Air, and Very Poor Air.

Depending on the detected air quality, the system provides appropriate visual and audible alerts, enabling users to understand environmental conditions instantly. The LCD displays real-time air quality information, while the LEDs and buzzer effectively notify users whenever pollution levels become hazardous.

The project demonstrates how embedded systems and IoT concepts can be applied to develop an economical, reliable, and user-friendly environmental monitoring solution. The successful simulation confirms the effectiveness of the proposed design and its suitability for educational and small-scale monitoring applications.

6.2 Future Scope

The proposed system can be further enhanced by incorporating advanced technologies and additional sensing capabilities. Some possible future improvements include:

- Integration with Wi-Fi or GSM modules for remote monitoring.
- Uploading sensor data to cloud platforms for real-time analysis.
- Development of a mobile application for live monitoring and notifications.
- Addition of temperature and humidity sensors for comprehensive environmental monitoring.
- Integration of GPS for location-based pollution mapping.
- Use of high-precision air quality sensors for improved measurement accuracy.
- Deployment in smart city infrastructure for large-scale environmental monitoring.

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